



Online

Object Oriented Programming
Computer Systems Engineering

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**Activity 1: Developing an application for a
basic shopping bill**

**Module 1: Object Oriented Programming
Languages Fundamentals**

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Code inserted into the given program

The original structure of the program was kept. The three sections marked with `//Insert your code here` were completed as follows, and each addition is commented in the source file.

1 · `setPrice(int category)`: unit price by product type

```
// Added code: set the unit price according to the product type
switch (category)
{
    case 1: price = 4.5; break;
    case 2: price = 3.5; break;
    case 3: price = 2.5; break;
    default: price = 0.0; break;    // not a valid product type
}
```

2 · `credit(double totalPrice)`: three months interest-free schedule

```
// Added code: print the remaining balance for each month of the credit
for (int month = 0; month <= 3; month++)
{
    System.out.printf("Balance payable of Month %d ---> %f\n", month, totalPrice - (month * partialPayment));
}
```

Note: the given code declared `parcial` but assigned `parcialPayment`; the declaration was renamed to `parcialPayment` so the file compiles. The change is commented in the source.

3 · `main(String args[])`: object creation

```
// Added code: create the Product object used to calculate the price and the credit
Product prod = new Product();
```

The program compiles and produces the expected output

```
● ● ●
$ javac Product.java
$ java Product
Which product would you like to order? 1, 2 or 3?
3
How many products would you like to order?
45
The total price of your order is: 112.500000
Balance payable of Month 0 --->112.500000
Balance payable of Month 1 --->75.000000
Balance payable of Month 2 --->37.500000
Balance payable of Month 3 --->0.000000
```

```
● ● ●
$ java Product
Which product would you like to order? 1, 2 or 3?
1
How many products would you like to order?
10
The total price of your order is: 45.000000
Balance payable of Month 0 --->45.000000
Balance payable of Month 1 --->30.000000
Balance payable of Month 2 --->15.000000
Balance payable of Month 3 --->0.000000
```

✓ Compiled with javac (no errors) and executed with the sample input of the assignment: the output matches the expected output line by line.